

Statistical Process Control for Monitoring the Particles With Excess Zero Counts in Semiconductor Manufacturing

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Abstract—In modern semiconductor manufacturing, one type of measured particle count data contains excess zeros, and the ratio of zeros in the measurements is usually larger than 50%. This type of particle count sample data cannot be well modeled by popular defect models such as Poisson, zero-inflated Poisson, generalized zero-inflated Poisson, and Neyman and Gamma-Poisson models. In this paper, a threshold-Poisson model was proposed to describe the particles with excess zero counts, and the method for parameter estimation was developed. Via comparison with those popular models by using 15 measured samples, it showed that the measurements are better modeled by the threshold-Poisson model. A control chart called threshold-c control chart was proposed and the control limits were derived. A reasonable minimum sample size for constructing control chart was also discussed based on simulations.

Index Terms—Control charts, excess zeros, particle, particle count, process control, quality control, statistical process control.

I. INTRODUCTION

AS THE chip density increases and semiconductor devices shrink in size, their yield, performance and reliability are sensitive to particulate contamination [1]. As a result, monitoring particles in semiconductor manufacturing is becoming more important. The particles in the air are measured and classified into discrete size using a particle counter, and in modern semiconductor manufacturing the particles can be measured by a laser particle counter. The commonly used method for monitoring the particle count is the c-chart [2]. The conventional c-chart is based on the assumption that the distribution of the number of defects is sufficiently well modeled by a Poisson distribution [2], [3]. In semiconductor manufacturing, we are usually concerned with the number of particles which are bigger than a setting value, causing the ratio of zeros in the

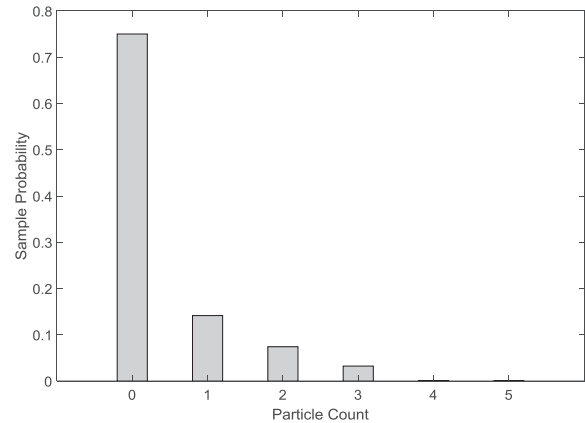


Fig. 1. An example of excess zeros in particle counts.

measurement to be usually larger than 50%. In this paper excess zero counts means the ratio of zeros is larger than 50%. Fig. 1 shows an example of the probabilities of the different particle counts. It illustrates that applying the traditional Poisson model is problematic and will lead to too many false alarms [4]–[7].

The distribution to describe the data with excess zero counts is defined as a zero-inflated distribution, and the most popular models are zero-inflated Poisson (ZIP) and generalized zero-inflated Poisson (GZIP). In addition, Neyman and Gamma-Poisson models are also used to describe the data with excess zero counts in recent years.

The ZIP model is proposed by Johnson and Kotz (1969). It assumes that the population consists of two types of measurements. The first type gives a Poisson-distributed count, which may be zero, whereas the second type always gives a zero count [4]. Xie and Goh (1993) [5] developed the SPC method based on the ZIP model, and proposed the original idea of the GZIP model informally at the end of his article. Böhning (1998) [6] found that these data such as the traffic accident data, criminal data and cancer registry data can be well modeled by the ZIP model. Xie *et al.* (2001) [7] investigated the use of the ZIP model and used it in SPC. Ridout *et al.* (2001) [8] provided a score test for testing the ZIP model. He *et al.* (2012) [9] proposed the CUSUM control chart based on the ZIP model. He *et al.* (2014) [10] proposed a combination of two CUSUM charts to detect the increases in both parameters of the ZIP process.

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The GZIP model was proposed by Chen *et al.* (2008), which is an extension of the ZIP model. The GZIP model assumes that the population consists of $n + 1$ ($n \geq 2$) types of measurements. The first n types give n different Poisson-distributed counts, which may be zero, whereas the last type always gives a zero count. Based on the GZIP model the Shewhart-type, CUSUM and ranked probability control chart were proposed by Chen *et al.* (2008) [11].

Dobbie and Welsh (2001) [12] modeled the data with excess zero counts by using the Neyman model, Dzupire *et al.* (2018) [13] described rainfall data with excess zero counts by using the Gamma-Poisson model.

In our opinion, there are two situations that would result in excess zero-defect phenomena. One is for “real defects”, that is the obtained defect count is the real defect count. In a ‘near zero-defect’ environment, it is understood that the individual product units are usually perfect, but there exists a random mechanism called shocks, which can lead to non-conformities in the sample whenever they are activated. The shocks occur randomly with a low probability so that a large number of zero nonconformity samples are observed [5]. The ZIP and GZIP models are proposed based on this situation.

The other situation resulting in excess zero-defect phenomena is for “measured defects”, that is, the defect count is the number of defects whose parameters are bigger than or equal to a setting value while the defects whose parameters are less than the setting value are ignored, leading to excess zero-defect phenomena. The zero defect here does not mean no defect but no defect bigger than the setting value. For example, in semiconductor manufacturing, if we only count the particles greater than a setting value in the clean room, a large number of zero particle counts will be observed. Zero particle count here does not mean no particle but no particle greater than the setting value. In practice, it is found that this kind of particle counts with excess zero counts cannot be well modeled by the popular models including Poisson, ZIP, GZIP, Neyman and Gamma-Poisson models.

In this paper, a model called threshold-Poisson model is proposed to describe this type of excess zero-defect phenomena, and the method for parameter estimation is developed. Via comparison with some other popular models by using 15 groups of measurements, it illustrates that the actual data is better modeled by this model. The control limits for the corresponding control chart named as threshold-c control chart are derived. The reasonable minimum sample size for constructing threshold-c control chart is discussed based on simulations.

The remaining parts of this paper are organized as follows: In Section II, we introduce the threshold-Poisson model, and develop the method for parameter estimation. Then the threshold-Poisson model is compared with some other popular models for defects. In Section III, the control limits for threshold-c control chart are derived, and some simulation results of sample size are given. In Section IV, we provide an illustrative example to explain the implementation of the threshold-c control chart. Finally, Section V summarizes the main work.

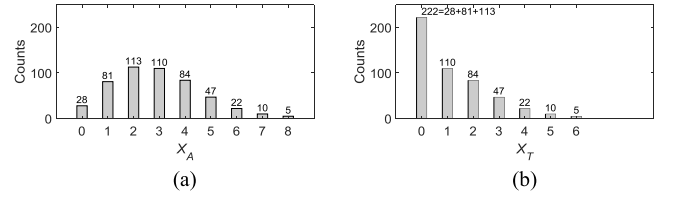


Fig. 2. An example illustrating the effect of measurement threshold.

II. THRESHOLD-POISSON MODEL FOR PARTICLES WITH EXCESS ZERO COUNTS

A. Threshold-Poisson Model

In semiconductor manufacturing the measured particle count is the number of the particles greater than a value set by a laser particle counter. This is equivalent to saying that there is a measurement threshold $T > 0$ in the particle counts. If the actual particle count X_A is bigger than the measurement threshold T , the particle count measured by particle counter will be $(X_A - T)$, and if not, the particle count measured by particle counter will be 0, leading to excess zero counts in the measured particle counts data. Therefore the measured particle counts X_T obtained from laser particle counter can be expressed as

$$X_T = \max(X_A - T, 0) \quad (1)$$

where X_A is the actual particle count, and X_T is the measured particle count obtained from a laser particle counter.

It is reasonable to assume that the actual particle count X_A follows the Poisson distribution, that is

$$p(X_A = m) = \frac{e^{-\lambda} \lambda^m}{m!} \quad m = 0, 1, 2, \dots \quad (2)$$

where $\lambda > 0$ is the parameter of the Poisson distribution.

According to Equations (1) and (2) the distribution of X_T can be expressed by Equation (3), which is defined as threshold-Poisson distribution in this paper.

$$p_m = p(X_T = m) = \begin{cases} \sum_{j=0}^T \frac{\lambda^j}{j!} e^{-\lambda}, & m = 0 \\ \frac{\lambda^{m+T}}{(m+T)!} e^{-\lambda}, & m = 1, 2, \dots \end{cases} \quad (3)$$

To illustrate the effect of the measurement threshold, a simple example is shown in Fig. 2. In this example, it is assumed that the measurement threshold $T = 2$, and the actual particle count X_A follows a Poisson distribution. Fig. 2 (a) shows the counts of actual particle count. Since $T = 2$, if X_A is 0, 1 or 2 then $X_T = 0$, so that the counts for $X_T = 0$ is $28 + 81 + 113 = 222$. If $X_A = 3$ then $X_T = 1$, so that the counts for $X_T = 1$ is 110. Similarly, the counts for $X_T = 2, 3, 4, 5$ and 6 correspond to the counts for $X_A = 4, 5, 6, 7$ and 8, respectively. Fig. 2 (b) shows the final result and the measurement threshold obviously results in too many zeros.

B. Parameter Estimation for Threshold-Poisson Model

It can be seen from Equation (3) that the theoretical probability of zero particles is

$$p_0 = \sum_{j=0}^T \frac{\lambda^j}{j!} e^{-\lambda}. \quad (4)$$

The cumulative distribution functions of the Poisson and the Chi-squared distributions are related in the following way [14]:

$$\sum_{j=0}^T \frac{\lambda^j}{j!} e^{-\lambda} = 1 - F_{\chi^2}(2\lambda; 2(T+1)) \quad (5)$$

where $F_{\chi^2}(2\lambda; 2(T+1))$ is the Chi-square cumulative distribution function at 2λ with $2(T+1)$ degrees of freedom. Since $p_0 = \sum_{j=0}^T \frac{\lambda^j}{j!} e^{-\lambda}$, then

$$\lambda = \frac{\chi_{p_0, 2(T+1)}^2}{2} \quad (6)$$

where $\chi_{p_0, 2(T+1)}^2$ is the upper p_0 percentage point of Chi-square distribution with $2(T+1)$ degrees of freedom.

In practice, we usually do not know λ and T , and they can be estimated from samples. Let k be the sample size, k_m be the counts of m in the sample, then the probability of m in the sample $p_{m, sample} = \frac{k_m}{k}$. Specially, the probability of 0 in the sample is $p_{0, sample}$. In case of excess zero counts, $p_{0, sample}$ is usually larger than 50%, and it is reasonable to replace p_0 with $p_{0, sample}$ in Equation (6) for parameter estimation, then

$$\lambda = \frac{\chi_{p_{0, sample}, 2(T+1)}^2}{2}. \quad (7)$$

There are two parameters in the threshold-Poisson model while there is only one mathematical relationship between λ and T , as shown in Equation (7). Therefore, a proper method for parameter estimation is proposed. The main idea of this method is to set the value of T , and then the corresponding value of λ can be obtained via Equation (7). After that the theoretical probabilities of each count value with T and λ can be calculated by Equation (3), and then the fit degree of theoretical probabilities and sample probabilities can be obtained. Setting different values of T , the corresponding λ and fit degrees can be obtained. The values of parameters T and λ with the best fit degree are selected as the estimated values of T and λ . In statistics, the determination coefficient, denoted by R^2 , provides a measure of how well observed outcomes are replicated by the model, based on the proportion of total variation of outcomes explained by the model [15]–[17]. Therefore, we chose to use the determination coefficient to determine the best fit values of parameters T and λ . The theoretical determination coefficient is

$$R^2(T) = \lim_{M \rightarrow \infty} \left[1 - \frac{\sum_{m=0}^M (p_m - p_{m, sample})^2}{\sum_{m=0}^M (p_{m, sample} - \bar{p}_{sample})^2} \right] \quad (8)$$

where $\bar{p}_{sample} = \frac{1}{M+1} \sum_{m=0}^M p_{m, sample}$.

In practice, the theoretical determination coefficient is not easily obtained. After thorough study, it is found that:

(1) There is a minimum cut off point M which does not affect the estimates of T and λ , and also is smaller than the largest count value, so that setting M equal to the largest count value would not affect the estimates of T and λ .

(2) While M is equal to the largest count value, the $R^2(T)$ is only a little smaller than the theoretical value and the relative

deviation is smaller than 0.001. That is, when M is equal to the largest count value, the $R^2(T)$ has a very high accuracy.

By above discussion, it is reasonable and convenient to choose M equal to the largest count value of the samples, and then the Equation (8) can be simplified as

$$R^2(T) = 1 - \frac{\sum_{m=0}^M (p_m - p_{m, sample})^2}{\sum_{m=0}^M (p_{m, sample} - \bar{p}_{sample})^2} \quad (9)$$

where M is the largest count value of the sample.

We found that with T increasing, the determination coefficient increases to a maximum value and then decreases. Therefore, the steps of parameter estimation are as follows:

Step 1: Initial condition preparation.

Step 1.1: Obtain the largest count value M of the samples, and compute the probability of m in sample $p_{m, sample}$, $m = 0, 1, 2, \dots, M$.

Step 1.2: Set initial value of T , $T = 0$.

Step 2: Compute the determination coefficient $R^2(T)$;

Step2.1: Compute the corresponding value of λ by substituting the value of T and the probability $p_{0, sample}$ into Equation (7).

Step 2.2: Compute the model probabilities p_m , ($m = 0, 1, 2, \dots, M$) by substituting the value of T and λ into Equation (3).

Step 2.3: Compute the determination coefficient $R^2(T)$ by substituting $p_{m, sample}$, p_m , and M into Equation (9).

Step 3: Similar to Step2, the determination coefficient $R^2(T+1)$ can be obtained.

Step 4: If $R^2(T+1) > R^2(T)$, then let $T := T + 1$ and go to Step 2. Otherwise the values of T and λ are selected as the estimated parametric value.

In our work, the parameter estimation based on the principle of maximum likelihood estimation (MLE) is also studied, and the relationship between T and λ is

$$p_{0, sample} \frac{\sum_{j=0}^{T-1} \frac{\lambda^j}{j!}}{\sum_{j=0}^T \frac{\lambda^j}{j!}} + \frac{E(Y)}{\lambda} + \frac{T(1 - p_{0, sample})}{\lambda} - 1 = 0 \quad (10)$$

where $E(Y)$ is the average of the sample. If Equation (7) is replaced by Equation (10) for parameter estimation, with the similar steps of parameters estimation, the estimated value of T and λ could be obtained, then the corresponding R^2 can be also obtained. The simulation result shows that the MLE method for parameter estimation is not as good as the above method, because the average of R^2 by MLE is a little smaller and the standard deviation of R^2 is a little larger than that by the above method. The MLE method will not be discussed in detail.

C. The Advantage of Threshold-Poisson Model in Practical Application

15 Lasair II-100 particle counters labeled as A1~A15 are placed in 15 different positions of a semiconductor clean room to measure the particle count in the air, and the sample data are shown in the Table I.

The parameter estimation method proposed in Section II-B is applied to each sample data, and the results are shown in the last two columns of the Table I.

TABLE I
SAMPLE DATA IN A REAL CLEAN ROOM

Label	Sample size	k_m (Counts of m)	$p_{0,sample}$	T	λ
A-1	1178	$k_0=1067, k_1=73, k_2=26, k_3=11, k_4=1$	0.9058	4	2.3869
A-2	1176	$k_0=880, k_1=166, k_2=87, k_3=38, k_4=1, k_5=1, k_{10}=1, k_{11}=1$	0.7489	3	2.5403
A-3	1077	$k_0=793, k_1=129, k_2=84, k_3=63, k_4=3, k_5=2, k_6=2, k_{12}=1$	0.7363	5	4.3022
A-4	1074	$k_0=959, k_1=86, k_2=24, k_3=5$	0.8929	2	1.1367
A-5	684	$k_0=620, k_1=40, k_2=17, k_3=6, k_4=1$	0.9064	4	2.3815
A-6	684	$k_0=506, k_1=84, k_2=49, k_3=40, k_4=3, k_5=1, k_{10}=1$	0.7398	5	4.2813
A-7	670	$k_0=588, k_1=53, k_2=19, k_3=10$	0.8776	3	1.8829
A-8	668	$k_0=500, k_1=79, k_2=54, k_3=32, k_4=3$	0.7485	5	4.2283
A-9	303	$k_0=186, k_1=65, k_2=27, k_3=17, k_4=4, k_5=0, k_6=1, k_7=1, k_8=1, k_9=1$	0.6139	2	2.2331
A-10	242	$k_0=167, k_1=28, k_2=23, k_3=9, k_4=5, k_5=1, k_6=2, k_7=2, k_8=1, k_{10}=2, k_{13}=1, k_{21}=1$	0.6901	6	5.4742
A-11	239	$k_0=164, k_1=36, k_2=17, k_3=7, k_4=8, k_5=2, k_6=2, k_7=1, k_{11}=1, k_{30}=1$	0.6862	4	3.7053
A-12	235	$k_0=139, k_1=45, k_2=26, k_3=7, k_4=6, k_5=4, k_6=2, k_9=1, k_{12}=1, k_{17}=1, k_{19}=1, k_{20}=1, k_{40}=1$	0.5915	2	2.3172
A-13	219	$k_0=136, k_1=37, k_2=25, k_3=6, k_4=5, k_5=4, k_6=1, k_8=1, k_9=1, k_{10}=1, k_{13}=1, k_{19}=1$	0.6210	3	3.1172
A-14	213	$k_0=137, k_1=34, k_2=19, k_3=9, k_4=6, k_5=1, k_{10}=1, k_{11}=1, k_{12}=1, k_{13}=1, k_{26}=1, k_{27}=1, k_{51}=1$	0.6432	3	3.018
A-15	212	$k_0=140, k_1=37, k_2=14, k_3=5, k_4=6, k_5=1, k_6=2, k_8=3, k_9=2, k_{10}=1, k_{14}=1$	0.6604	2	2.0602

The Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC) are widely used in the model selection. The smaller the AIC and BIC values are, the better the model is.

The AIC and BIC of the model are formally defined as [16], [17]

$$AIC = 2K - 2 \ln(L) \quad (11)$$

$$BIC = K \ln(n) - 2 \ln(L) \quad (12)$$

where K is the number of estimated parameters in the model, L is the likelihood function for the model, and n is the number of sample data. After parameter estimation, Equations (11) and (12) can be expressed as

$$AIC = 2K - 2k_i \sum_{i=1}^M \ln p_i \quad (13)$$

$$BIC = K \ln(n) - 2k_i \sum_{i=1}^M \ln p_i \quad (14)$$

where p_i is the modeled probability of count i of the corresponding model.

After calculation, the determination coefficient, AIC and BIC values of all models are shown in Table II, III and IV.

Table II suggests that: (1) The R^2 of the threshold-Poisson for each sample is equal or larger than that of other models. (2) The average of R^2 of threshold-Poisson is the largest and the standard deviation of R^2 is the smallest. (3) The smallest R^2 of threshold-Poisson is 0.9954, which means that all the particle data of A1-A15 can be well modeled by threshold-Poisson model.

From Table III and IV, it is observed that: (1) The AIC and BIC of threshold-Poisson model are smaller than those of the Poisson model. (2) The AIC and BIC of the GZIP($n = 2$), GZIP($n = 3$) and gamma-Poisson for A10-A15 are smaller than those of threshold-Poisson. This situation is caused by the abnormally large values. For example, if the maximums

of A10-A15 are deleted, then nearly half of the results will be reversed. (3) For about half of A1-A9, the AIC and BIC of the ZIP and Neyman are smaller than those of the threshold-Poisson. (4) The AIC and BIC of the threshold-Poisson are smaller than those of the popular defect models in most cases.

By above, the determination coefficient, AIC and BIC all suggest that the threshold-Poisson model has better performance for particles with excess zero counts than the other popular models.

Note that, the GZIP(n) means that the process contains n kinds of shocks and $2n$ parameters, and the parameter estimation method is given in [11].

Taking the data from 'A-3' and 'A-9' as examples to explicitly show that the threshold-Poisson model can fit the particle data with excess zeros well and stably, the results are shown in Fig. 3. Fig. 3 (a) shows that the threshold-Poisson, ZIP and Neyman modeled values are very close to the observed values of A-3 on the whole, while Fig. 3 (b) shows that the threshold-Poisson and Gamma-Poisson modeled values are very close to the observed values of A-9 on the whole, which means that the particle data with excess zeros can be well and stably modeled by the threshold-Poisson model. Fig. 3 also shows that it is difficult to model all the observed values accurately.

III. CONTROL CHART BASED ON THRESHOLD-POISSON MODEL (THRESHOLD-C CONTROL CHART)

A. Control Limits for Threshold-c Control Chart

In this paper, the corresponding control chart based on the threshold-Poisson model is named the threshold-c control chart.

The ratio of zeros of the particle data discussed in this paper is usually larger than 50%, so that the lower control limit LCL_T for the threshold-c control chart is always zero, that is, $LCL_T = 0$.

TABLE II
 R^2 OF DIFFERENT MODELS FOR REAL WORLD DATA

Label	Threshold-Poisson	Poisson	ZIP	GZIP (n=2)	GZIP (n=3)	Neyman	Gamma-poisson
A-1	1.0000	0.9923	1.0000	0.9981	0.9981	1.0000	0.9996
A-2	0.9997	0.9451	0.8159	0.9872	0.9872	0.9995	0.9974
A-3	0.9986	0.8842	0.9983	0.9915	0.9915	0.9983	0.9895
A-4	1.0000	0.9957	1.0000	0.9962	0.9962	1.0000	0.9998
A-5	1.0000	0.9908	1.0000	0.9984	0.9984	1.0000	0.9995
A-6	0.9986	0.8891	0.9982	0.9908	0.9908	0.9981	0.9901
A-7	0.9999	0.9860	0.9999	0.9972	0.9972	0.9998	0.9992
A-8	0.9989	0.8945	0.9975	0.9929	0.9929	0.9930	0.9826
A-9	0.9974	0.8899	0.9510	0.9557	0.9565	0.9789	0.9986
A-10	0.9986	0.6552	0.8997	0.9916	0.9916	0.9145	0.9947
A-11	0.9981	0.7304	0.8354	0.9839	0.9839	0.8445	0.9713
A-12	0.9970	0.5814	0.5813	0.9681	0.9681	0.5948	0.8909
A-13	0.9971	0.7075	0.8094	0.9759	0.9759	0.8434	0.9833
A-14	0.9987	0.4642	0.6962	0.9832	0.9832	0.7010	0.9087
A-15	0.9954	0.7187	0.8747	0.9738	0.9738	0.8987	0.9939
Average of R^2	0.9985	0.8217	0.8972	0.9856	0.9857	0.9176	0.9799
Standard deviation of R^2	0.0014	0.1677	0.1294	0.0124	0.0123	0.1254	0.0337

TABLE III
AIC OF DIFFERENT MODELS FOR REAL WORLD DATA

Label	Threshold-Poisson	Poisson	ZIP	GZIP (n=2)	GZIP (n=3)	Neyman	Gamma-poisson
A-1	940.52	1052.63	938.83	970.96	974.96	942.42	945.82
A-2	1980.93	2168.48	1995.28	2048.08	2052.08	1985.57	1990.55
A-3	1976.83	2265.15	1972.02	2008.55	2012.55	1966.83	1999.53
A-4	894.16	941.80	893.12	1035.22	1039.22	893.89	897.10
A-5	547.96	624.44	549.10	642.25	646.25	549.58	554.70
A-6	1239.67	1413.71	1235.53	1292.95	1296.95	1233.83	1252.97
A-7	652.22	722.35	653.51	759.09	763.09	652.49	657.63
A-8	1157.24	1284.82	1142.76	1227.66	1231.66	1154.29	1175.62
A-9	717.01	783.01	742.95	744.89	734.57	717.65	725.88
A-10	624.09	868.03	800.50	587.26	591.26	723.99	577.80
A-11	628.90	818.86	984.22	599.38	594.57	883.96	563.83
A-12	953.53	1142.07	1788.36	683.51	687.51	1579.48	698.63
A-13	634.90	781.83	794.85	610.48	614.48	708.31	574.23
A-14	938.87	1224.97	2178.30	602.96	606.96	1986.25	609.39
A-15	608.18	731.71	673.38	587.86	591.86	616.77	518.49

It is shown from Equation (1) that the upper control limit UCL_T for the threshold-c chart with parameters λ and T should be T less than the upper control limit UCL_0 for the traditional c chart with parameter λ . According to probability theory and statistics, the Poisson distribution is similar to the normal distribution only when λ is large enough, such as larger than 10. Table I suggests that λ is usually much smaller than 10 for particles in a clean room, which means that the corresponding Poisson distribution is far from the normal distribution. So, the UCL_0 for c charts need to be modified. Winterbottom (1993) proposed a modified 3σ upper control limit based on Cornish–Fisher expansion [18], [19], that is

$$UCL_0 = \lambda + 3\sqrt{\lambda} + 4/3. \quad (15)$$

The modified 3σ upper control limit UCL_0 is the (almost) exact 0.99865 upper control limit [18] [19].

Therefore, the upper control limit UCL_T for the threshold-c control chart should be

$$UCL_T = \lambda + 3\sqrt{\lambda} + 4/3 - T. \quad (16)$$

Then, the UCL_T for threshold-c control chart is also the (almost) exact 0.99865 upper control limit, so type I error is always around 0.00135.

B. The Minimum Sample Size Needed to Construct the Threshold-c Control Chart

Shewhart control charts are usually constructed using 20~30 initial samples. The ratio of zeros of the particle data discussed in this paper is usually larger than 50%. Obviously, if the sample size is still 20~30, the inherent (common-cause) variation can have a significant impact on the sample probabilities of non-zero particle counts, which

TABLE IV
BIC OF DIFFERENT MODELS FOR REAL WORLD DATA

Label	Threshold-Poisson	Poisson	ZIP	GZIP (n=2)	GZIP (n=3)	Neyman	Gamma-poisson
A-1	950.67	1062.77	948.98	991.25	1005.39	952.57	955.96
A-2	1991.07	2178.62	2005.41	2068.35	2082.49	1995.71	2000.69
A-3	1986.79	2275.11	1981.98	2028.48	2042.44	1976.79	2009.49
A-4	904.12	951.76	903.07	1055.14	1069.10	903.85	907.06
A-5	557.02	633.49	558.16	660.37	673.42	558.64	563.75
A-6	1248.73	1422.77	1244.59	1311.06	1324.12	1242.89	1262.03
A-7	661.24	731.37	662.53	777.12	790.13	661.50	666.64
A-8	1166.25	1293.82	1151.77	1245.68	1258.69	1163.30	1184.63
A-9	724.44	790.44	750.38	759.75	756.85	725.08	733.30
A-10	631.07	875.01	807.48	601.22	612.19	730.96	584.78
A-11	635.85	825.81	991.17	613.29	615.43	890.91	570.78
A-12	960.45	1148.99	1795.28	697.35	708.27	1586.40	705.55
A-13	641.68	788.61	801.62	624.04	634.82	715.09	581.00
A-14	945.59	1231.70	2185.02	616.41	627.13	1992.98	616.11
A-15	614.89	738.42	680.10	601.28	612.00	623.48	525.20

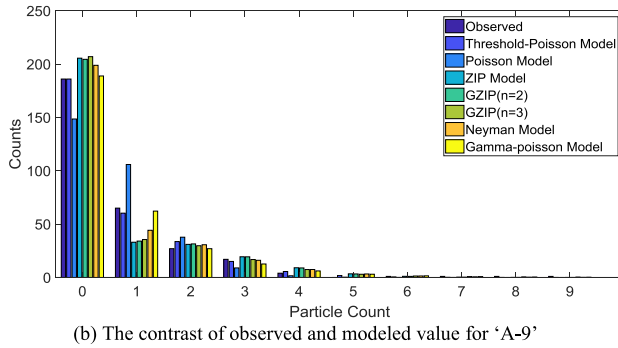
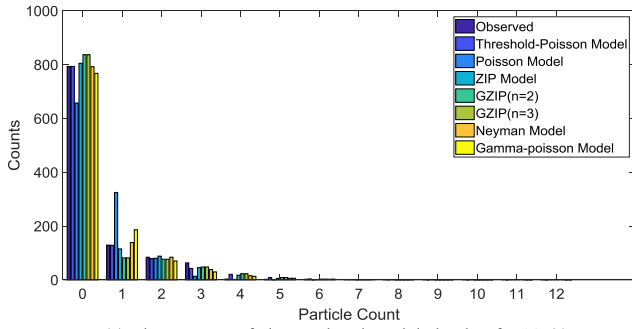


Fig. 3. The contrast of observed and modeled value.

affects the calculating precision of UCL_T . Therefore, more initial samples are needed to construct the threshold-c control chart. In practice, a laser particle counter usually measures the particle count once per minute, and then 1,440 samples can be gathered one day. So, a reasonably large initial sample size such as several hundred initial samples is acceptable.

In order to determine the minimum initial sample size, lots of simulation experiments are carried out by the following steps:

Step 1: Set the values of p_0 and T , and compute the corresponding value of λ by Equation (7), then compute UCL_{TT} by Equation (16), where UCL_{TT} is the theoretical upper control limit.

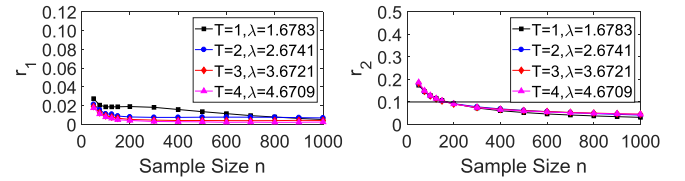


Fig. 4. Simulation results with $p_0 = 0.5$.

Step 2: Set the value of n , and generate 10^6 groups of the Poisson distribution random numbers with parameter λ , and each group contains n numbers. Then obtain the threshold-Poisson distribution random numbers by substituting the random numbers and T value into Equation (1).

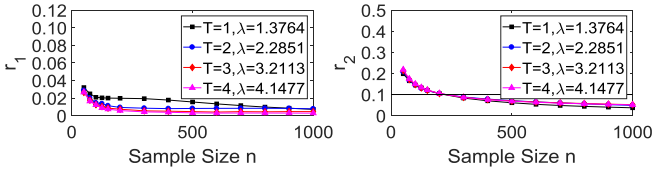
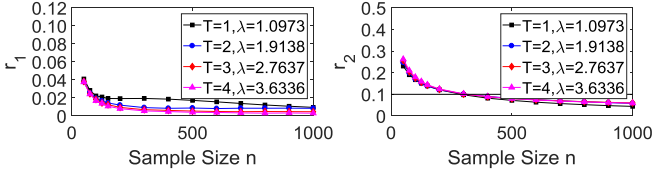
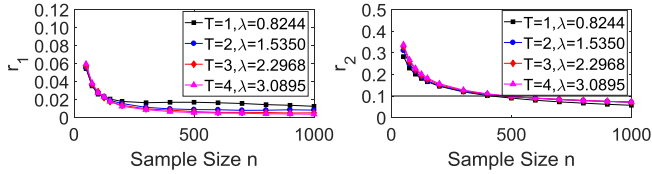
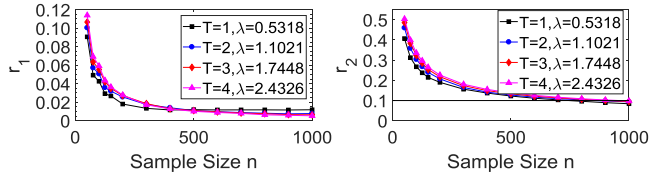
Step 3: Estimate T and λ for each group of random numbers using the parameter estimation method proposed in Section II-B, and then 10^6 groups of estimated values of T and λ can be obtained.

Step 4: Compute the UCL_T for each group by substituting corresponding estimated values T and λ into Equation (16), and then we obtain 10^6 different UCL_T .

Step 5: Compute the mean and standard deviation of the 10^6 UCL_T , and they are represented as $\text{mean}(UCL_T)$ and $\text{std}(UCL_T)$. Compute the ratio $r_1 = (\text{mean}(UCL_T) - UCL_{TT})/UCL_{TT}$ and $r_2 = \text{std}(UCL_T)/UCL_{TT}$. That is, r_1 indicates the relative deviation between the expectation of UCL_T obtained by the parameter estimation method and the theoretical value UCL_{TT} ; and r_2 indicates the relative dispersion between the dispersion of UCL_T and the theoretical value UCL_{TT} .

Some simulation results are shown in Fig. 4~8.

(1) It is considered reasonable that if $r_2 \leq 0.1$, the dispersion of UCL_T is acceptable in engineering application. From Fig. 4 (a), Fig. 5 (a), Fig. 6 (a), Fig. 7 (a) and Fig. 8 (a), the minimum sample size needed to construct the threshold-c control chart (while $r_2 = 0.1$) for different p_0 can be obtained as shown in Table V. Table V suggests that the minimum sample size increases with the increase of p_0 .

Fig. 5. Simulation results with $p_0 = 0.6$.Fig. 6. Simulation results with $p_0 = 0.7$.Fig. 7. Simulation results with $p_0 = 0.8$.Fig. 8. Simulation results with $p_0 = 0.9$.

(2) Table V also shows the corresponding r_1 , and they are very close to 0, which means that the calculated UCL_T is almost unbiased.

(3) As seen from Table V, the non-zero number of 90 is enough for different p_0 .

In summary, to construct a threshold-c control chart, about 90 non-zero sample data needs to be collected, or the minimum expected sample size n_0 is about $90/(1 - p_0)$. Then in practice, the method of data collection for constructing the threshold-c control chart is: collect particle counts until about 90 non-zero particle counts are collected using a particle counter. It is easy to know that the sample size n obtained by this method follows a negative Binomial distribution. After study, it turns out that the probability of n falling inside the interval of $n_0[1 \pm (0.1 + 0.1p_0)]$ is about 95%. The maximum relative deviation is less than 20%, and it is acceptable in engineering application.

C. The Steps of Establishing Threshold-c Control Chart

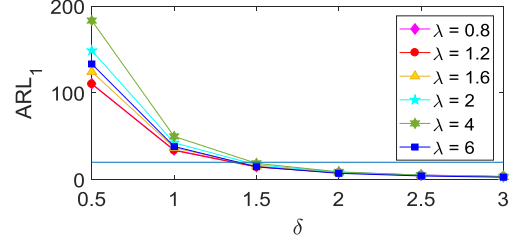
Through the above analysis and discussions, the steps of constructing a threshold-c control chart are as follows:

Step 1: Collect particle counts until about 90 non-zero particle counts are collected using a particle counter.

Step 2: Estimate T and λ using the parameter estimation method proposed in Section II-B.

TABLE V
THE MINIMUM SAMPLE SIZE FOR DIFFERENT p_0

p_0	0.5	0.6	0.7	0.8	0.9
sample size	150	225	300	450	900
Counts of nonzero	75	90	90	90	90
r_1	≤ 0.02	≤ 0.02	≤ 0.02	≤ 0.02	≤ 0.015

Fig. 9. The out-of-control average run length (ARL_1) for the proposed threshold-c control chart.

Step 3: Compute the UCL_T by substituting estimated value of T and λ into Equation (16).

Step 4: Draw the measured particle count and UCL_T in the threshold-c control chart.

D. The Performance for Threshold-c Control Chart

If λ shifts to another value of $\lambda + \delta\sqrt{\lambda}$, the probability of not detecting this shift on the first subsequent sample, or the type II error β , is

$$\begin{aligned}
 \beta &= p(X_T < UCL_T) \\
 &= p(X_T = 0) + p(X_T = 1) + \dots + p(X_T = \lfloor UCL_T \rfloor) \\
 &= \left[\sum_{j=0}^T p(X_A = j) \right] + p(X_A = 1 + T) + \dots \\
 &\quad + p(X_A = \lfloor UCL_T \rfloor + T) \\
 &= \sum_{j=0}^{\lfloor UCL_T \rfloor + T} \frac{(\lambda + \delta\sqrt{\lambda})^j}{j!} e^{-(\lambda + \delta\sqrt{\lambda})} \\
 &= \sum_{j=0}^{\lfloor UCL_0 \rfloor} \frac{(\lambda + \delta\sqrt{\lambda})^j}{j!} e^{-(\lambda + \delta\sqrt{\lambda})} \quad (17)
 \end{aligned}$$

where $\lfloor x \rfloor$ means the nearest integer less than or equal to x . Then the average run length of the proposed threshold-c control chart when particle count is out of control (called ARL_1) is

$$ARL_1 = \frac{1}{1 - \beta}. \quad (18)$$

Fig. 9 presents a set of curves for ARL_1 , and it shows that the $ARL_1 < 60$ while $\delta = 1$ and $ARL_1 < 20$ while $\delta = 1.5$. The particle counter can collect the particle count once a minute, so that it is acceptable in practice. To detect the small shift more quickly, the corresponding CUSUM or EWMA control charts based on threshold-Poisson model should be used.

E. The CUSUM and EWMA Charts

The threshold-c chart is a Shewhart control chart, and the major disadvantage of the Shewhart control chart is

TABLE VI
SAMPLE DATA COLLECTED IN A CLEAN ROOM

NO	particle count	NO	particle count	NO	particle count	NO	particle count	NO	particle count
1	0	51	0	101	1	151	0	201	0
2	0	52	0	102	0	152	0	202	0
3	2	53	2	103	0	153	0	203	0
4	1	54	2	104	0	154	1	204	2
5	0	55	4	105	0	155	0	205	0
6	0	56	0	106	0	156	0	206	0
7	2	57	0	107	0	157	0	207	0
8	1	58	0	108	1	158	0	208	2
9	2	59	0	109	3	159	0	209	0
10	0	60	2	110	0	160	0	210	1
11	2	61	0	111	1	161	1	211	0
12	1	62	0	112	0	162	0	212	0
13	0	63	0	113	0	163	0	213	0
14	0	64	0	114	3	164	1	214	0
15	1	65	0	115	2	165	3	215	1
16	3	66	0	116	2	166	0	216	0
17	1	67	0	117	3	167	1	217	0
18	0	68	0	118	0	168	1	218	0
19	0	69	3	119	0	169	1	219	0
20	0	70	2	120	1	170	0	220	3
21	0	71	0	121	0	171	0	221	0
22	0	72	0	122	0	172	0	222	0
23	0	73	0	123	1	173	1	223	0
24	0	74	0	124	2	174	1	224	0
25	2	75	1	125	1	175	0	225	2
26	1	76	1	126	0	176	0	226	0
27	0	77	2	127	0	177	0	227	0
28	0	78	0	128	4	178	0	228	1
29	1	79	0	129	2	179	0	229	2
30	1	80	1	130	1	180	0	230	1
31	2	81	0	131	0	181	2	231	0
32	1	82	0	132	0	182	0	232	0
33	0	83	1	133	0	183	5	233	1
34	0	84	1	134	0	184	0	234	0
35	1	85	0	135	0	185	0	235	1
36	0	86	2	136	1	186	0	236	0
37	0	87	1	137	0	187	0	237	0
38	0	88	2	138	0	188	0	238	0
39	1	89	2	139	0	189	0	239	1
40	0	90	0	140	0	190	0	240	0
41	2	91	0	141	1	191	0	241	0
42	0	92	3	142	1	192	1	242	0
43	1	93	3	143	0	193	0	243	0
44	0	94	0	144	1	194	0	244	1
45	0	95	0	145	3	195	2	245	0
46	0	96	0	146	0	196	0	246	1
47	0	97	0	147	0	197	0	247	0
48	1	98	0	148	0	198	0	248	2
49	0	99	1	149	2	199	0	249	3
50	0	100	0	150	0	200	0	250	0

that it is relatively insensitive to small process shifts, such as, on the order of about 1.5σ or less [3]. Since the methods of establishing EWMA and CUSUM charts based on the threshold-Poisson model are basically the same as those based on the other models, the method and key points of establishing EWMA and CUSUM charts

TABLE VII
STATISTICAL RESULTS OF THE SAMPLE DATA

Particle count(m)	0	1	2	3	4	5
Counts(k_m)	160	49	27	11	2	1
Sample probability ($p_{m,sample}$)	0.640	0.196	0.108	0.044	0.008	0.004

based on the threshold-Poisson model are only explained briefly.

1) *The CUSUM Chart Based on the Threshold-Poisson Model*: For the characteristic of the threshold-Poisson model, when the value set by the laser particle is not changed, the measurement threshold T will be constant. If the CUSUM is designed to detect a shift from λ to λ_1 , the CUSUM statistic B_i is updated according to the following

$$B_i = \max(0, B_{i-1} + K_i) \quad (19)$$

where $B_0 = 0$, and K_i is based on the log-likelihood ratio for a specified shift size, i.e.,

$$K_i = \begin{cases} \ln \frac{\sum_{j=0}^T \frac{\lambda_1^j}{j!} e^{-\lambda_1}}{\sum_{j=0}^T \frac{\lambda^j}{j!} e^{-\lambda}}, & X_{Ti} = 0 \\ \ln \frac{\frac{\lambda^{X_{Ti}+T}}{(X_{Ti}+T)!} e^{-\lambda}}{\frac{\lambda_1^{X_{Ti}+T}}{(X_{Ti}+T)!} e^{-\lambda_1}}, & X_{Ti} > 0 \end{cases} \quad (20)$$

where the X_{Ti} is the i th measurement.

The CUSUM chart signals when $B_i > h$, where h is the upper control limit, which can be obtained through simulation to achieve the desired in-control performance. And the approximate relationship between the h and the α error is $h = -\ln \alpha$ [11].

2) *The EWMA Chart Based on the Threshold-Poisson Model*: The EWMA statistic Z_i is updated according to the following

$$Z_i = rX_{Ti} + (1 - r)Z_{i-1} \quad (21)$$

where $0 < r < 1$ is a constant and $Z_0 = \mu_0$, μ_0 is the theoretical mean with the parameters T and λ , and μ_0 can be obtained via the following equation (for more detail derivation see the Appendix)

$$\mu_0 = \lambda - T + \sum_{m=0}^T (T - m) \frac{\lambda^m}{m!} e^{-\lambda}. \quad (22)$$

The control limits for the EWMA control charts are as follows.

$$\begin{aligned} UCL &= \mu_0 + L\sigma \sqrt{\frac{r}{2-r} [1 - (1-r)^{2i}]} \\ LCL &= \mu_0 - L\sigma \sqrt{\frac{r}{2-r} [1 - (1-r)^{2i}]} \end{aligned} \quad (23)$$

where σ is the theoretical standard deviation with the parameters T and λ . And σ can be obtained via the following equation (for more detail derivation see the Appendix)

$$\sigma = \sqrt{\lambda - \sum_{m=0}^T (2\lambda - T - m)(T - m) \frac{\lambda^m}{m!} e^{-\lambda} - \left(\sum_{m=0}^T (T - m) \frac{\lambda^m}{m!} e^{-\lambda} \right)^2} \quad (24)$$

TABLE VIII
THE RESULTS OF ITERATIVE PROCESS

Threshold(T)	0	1	2	3
λ	0.4463	1.2630	2.1357	3.0323
$R^2(T)$	0.9619	0.9958	0.9996	0.9966

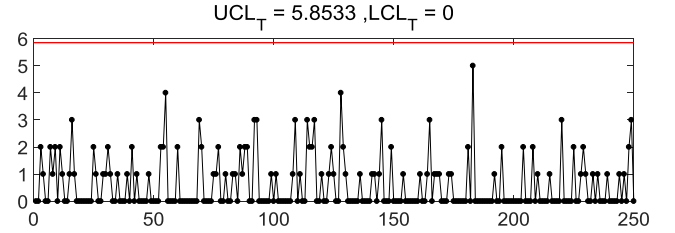


Fig. 10. An example using the threshold-c control chart to monitor particles with excess zero counts.

where L is the width of the control limits, and can be obtained through a Monte Carlo approach to achieve the desired in-control performance.

IV. AN APPLICATION IN REAL WORLD

According to Section III-C, an example is given to implement SPC (with the threshold-c chart) to monitor particles with excess zero counts in a clean room.

Step 1: In a clean room, we count the particles greater than a setting value using a Lasair II-100 particle counter, collect the particle count once a minute, and get 250 particle counts as initial sample data, which contains 90 non-zero particle counts. The sample data are shown in Table VI.

Step 2: Estimate T and λ using the parameter estimation method proposed in Section II-B.

From Table VI, the counts of different particles are shown in Table VII.

Then estimate T and λ using the parameter estimation method proposed in Section II-B, and the results of the iterative process are shown in Table VIII.

Table VIII suggests that the determination coefficient increases with the increase of T from 0 to 2, and then decreases with the further increase of T . And with the threshold $T=2$, the corresponding $\lambda = 2.1357$. The threshold-Poisson fits best with the observed measurement data, since $R^2 = 0.9996$ is very close to 1. So the parameter estimation results are $T = 2$ and $\lambda = 2.1357$.

Step 3: By substituting $T = 2$ and $\lambda = 2.1357$ in Equation (16), the upper control limits for the threshold-c control chart: $UCL_T = 5.8533$, can be obtained.

Step 4: After drawing the measured particle count and UCL_T in the threshold-c control chart, the threshold-c chart is obtained as shown in Fig. 10.

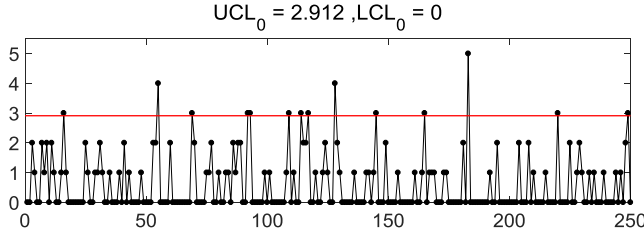


Fig. 11. Using c control chart to monitor the particle with excess zero counts.

Fig. 10 suggests that no point falls outside the UCL_T , which means that the variation of particle count in the air is in control.

As a contrast, if we choose the traditional c chart to monitor this process directly using the initial 250 samples, then we can get the c control chart as shown in Fig. 11.

Fig. 11 suggests that 14 points fall outside the UCL , which indicates that the process is seriously out of control. The result comes from misusing the c control chart.

V. CONCLUSION

In this paper, the statistical process control for monitoring particles with excess zero counts in semiconductor manufacturing is studied. The threshold-Poisson model is proposed and a suitable parameter-estimation method is developed. Then, by using 15 groups of measured particle data with excess zeros, the determination coefficient, AIC and BIC all suggest that threshold-Poisson model has better performance than the other popular models. After that, a control chart named the threshold-c control chart is proposed and the control limits are derived. Next the minimum sample size needed to construct the threshold-c control chart is discussed. The methods and key points for establishing the EWMA and CUSUM charts based on the threshold-Poisson model are also given. Finally, we use the threshold-c control chart to monitor the particle count, and the result is reasonable. Meanwhile, the determination coefficient is very close to 1, which illustrates the correctness of the model and the parameter estimation method. As a contrast, if we choose traditional c control chart to monitor the particle count, there would be too many false alarms.

APPENDIX

The derivation of Equations (22) and (24) based on the threshold-Poisson model

The theoretical mean μ_0 is deduced as follow:

$$\begin{aligned}\mu_0 &= \sum_{m=1}^{+\infty} m \frac{\lambda^{m+n}}{(m+T)!} e^{-\lambda} \\ &= \sum_{m=1}^{+\infty} (m+T) \frac{\lambda^{m+n}}{(m+T)!} e^{-\lambda} - T \sum_{m=1}^{+\infty} \frac{\lambda^{m+T}}{(m+T)!} e^{-\lambda} \\ &= \sum_{m=n+1}^{+\infty} m \frac{\lambda^m}{m!} e^{-\lambda} - T \sum_{m=T+1}^{+\infty} \frac{\lambda^m}{m!} e^{-\lambda}\end{aligned}$$

$$\begin{aligned}&= \left\{ \left[\sum_{m=T+1}^{+\infty} m \frac{\lambda^m}{m!} e^{-\lambda} + \sum_{m=0}^T m \frac{\lambda^m}{m!} e^{-\lambda} \right] - \sum_{m=0}^T m \frac{\lambda^m}{m!} e^{-\lambda} \right\} \\ &\quad - n \left(1 - \sum_{m=0}^T \frac{\lambda^m}{m!} e^{-\lambda} \right) \\ &= \lambda - T + T \sum_{m=0}^T \frac{\lambda^m}{m!} e^{-\lambda} - \sum_{m=0}^T m \frac{\lambda^m}{m!} e^{-\lambda} \\ &= \lambda - T + \sum_{m=0}^T (T-m) \frac{\lambda^m}{m!} e^{-\lambda}\end{aligned}$$

Similarly, the second order moment $E(X_T^2)$ can be obtained

$$E(X_T^2) = \lambda - \sum_{m=0}^T (2\lambda - T - m)(T - m) \frac{\lambda^m}{m!} e^{-\lambda}$$

Then the theoretical standard deviation σ is

$$\begin{aligned}\sigma &= \sqrt{E(X_T^2) - \mu_0^2} \\ &= \sqrt{\lambda - \sum_{m=0}^T (2\lambda - T - m)(T - m) \frac{\lambda^m}{m!} e^{-\lambda} - \left(\sum_{m=0}^T (T - m) \frac{\lambda^m}{m!} e^{-\lambda} \right)^2}\end{aligned}$$

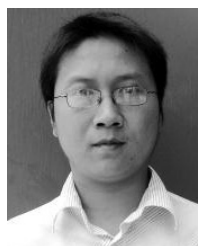
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